

Self-Maintained Order and Hysteretic Collapse in a Non-Equilibrium Rotational Lattice

Order-dependent degradation and repair as a source of nonlinear feedback, bistability, and active maintenance

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Abstract

We study the maintenance of directional order in a non-equilibrium rotational lattice: a two-dimensional array of interacting rotors whose ordering couplings are removed by an order-dependent flux and restored by repair. Because degradation accelerates with local orientational disorder, the stationary state is a non-equilibrium steady state (NESS). In an annealed mean-field reduction controlled by the ratio ρ of repair to shielded degradation, the feedback makes order bistable: for cooperativity $h > 1$ and gain above a computed threshold, two saddle-node bifurcations bound a hysteresis window, whereas without feedback there is one fixed point, $I^* = 1/(1 + \rho)$. Kinetic Monte Carlo tests this. When degradation senses only the nearest neighbours the lattice is not bistable: the hysteresis loop closes as the sweep slows, its area vanishing in the quasi-static limit for $L = 16\text{--}64$, and ordered and collapsed states converge throughout the window, because averaging the steep feedback over a broad distribution of local disorder flattens it. When degradation senses disorder over larger regions bistability returns; imposing the closure itself on the lattice gives a window that opens at the mean-field saddle-node. Two further results are established: an exact order–entropy bridge, $dS/dU_\chi = -\kappa$ for the maximum-entropy von Mises law, and finite-size scaling ($L = 16\text{--}96$) placing the field-free loss of stiffness at a diluted fraction $f_{KT} \approx 0.205$, far below site percolation (0.407), a Berezinskii–Kosterlitz–Thouless mechanism. Schnakenberg entropy production quantifies the cost of maintenance. Static hysteresis is predicted only when degradation senses order over an extended range.

Keywords: nonlinear dynamics; non-equilibrium statistical mechanics; bistability and hysteresis; Berezinskii–Kosterlitz–Thouless transition; entropy production; kinetic Monte Carlo

1. Introduction

Many physical systems maintain order not as a thermodynamic ground state but as a dynamical balance: a directionally ordered medium is continuously eroded by an external flux while a maintenance process restores it [1–3]. Order then persists or collapses according to a competition of rates. When the erosion depends on the current state of the medium, the balance is governed by feedback, which admits the multistability and the dissipation characteristic of non-equilibrium systems [4–6]. Instances include the orientational order of a polymer film under photodegradation with self-healing [7], the alignment of magnetic domains under thermally activated flips [8] with active error-correcting rewrites, and the directional order of a molecular monolayer under chemical attack with turnover [9].

The setting differs from the best-known orientational non-equilibrium systems. In flocking and active-matter models, driving generates and stabilizes order through self-propulsion and alignment, and long-range order can survive in two dimensions where the equilibrium XY model cannot [10, 11]; in the equilibrium XY model, order is lost through the Berezinskii–Kosterlitz–Thouless (BKT) transition [12]. Here order is neither generated by the drive nor protected by an equilibrium ground state: it is maintained against a state-dependent loss. Keeping information-bearing order away from equilibrium also has a thermodynamic price [13, 14], of which kinetic proofreading is the classic example in which dissipation buys fidelity [15], and which stochastic thermodynamics makes quantitative [16, 17].

We formulate this as a non-equilibrium interacting particle system [18] in which reversible rotor dynamics is coupled to a non-conservative, order-dependent site process. The coupling produces a nonlinear mean-field reduction that is bistable, with saddle-node bifurcations bounding a hysteresis window in which a maintained and a collapsed NESS coexist (Section 4.2), a behavior absent without the feedback (Section 4.1). We then test the reduction on the lattice by kinetic Monte Carlo designed to separate static from rate-dependent hysteresis (Section 4.3) and find that its bistability survives only when degradation senses order over an extended region; with nearest-neighbour sensing the lattice has a single NESS, and we identify why (Section 4.4). Three further contributions support the picture: an exact bridge between a directional order parameter and configurational entropy (Section 3); a finite-size-scaling analysis locating the field-free transition relative to site percolation and identifying a BKT mechanism (Section 5); and the Schnakenberg entropy production rate quantifying the thermodynamic cost of maintenance (Section 6). The order-dependence of the degradation, a site in a disordered neighborhood degrading faster, is what makes the stationary state a genuine NESS rather than a product of an uncoupled two-state chain and an equilibrium rotor gas, and is the source of the nonlinear phenomenology.

61 2. The model

62 Let $\Lambda = \{1, \dots, L\}^2$ be a square lattice with $N = L^2$ sites and periodic boundaries. Each site i carries a rotor
 63 orientation $\theta_i \in [0, 2\pi)$ and a binary integrity $g_i \in \{1, g_0\}$, $0 < g_0 \ll 1$. The energy is

$$64 \quad H(\theta, g) = -J_0 \sum_{\langle ij \rangle} g_i g_j \cos(\theta_i - \theta_j) - h_0 \sum_i g_i \cos \theta_i. \quad (1)$$

65 The dynamics interleaves a reversible rotor relaxation with a non-conservative, order-dependent site
 66 process.

67 *Reversible rotor relaxation.* At fixed g , the orientations evolve by a Metropolis kernel reversible with
 68 respect to the Gibbs measure $\propto e^{-H/T}$ ($k_B T = 1$); alone, this equilibrates the rotors in the quenched integrity
 69 landscape and satisfies detailed balance.

70 *Order-dependent site process.* Each intact site ($g_i = 1$) is *diluted* ($g_i \rightarrow g_0$) at a rate that depends on its
 71 local orientational disorder,

$$72 \quad \lambda_i = \lambda_0 \left[1 + \beta \frac{\delta_i^h}{K^h + \delta_i^h} \right] \quad (2)$$

$$73 \quad \delta_i = 1 - \left| \frac{1}{4} \sum_{j \in \partial i} e^{i\theta_j} \right| \quad (3)$$

$$74 \quad \lambda_0 = k G (1 - s u_s). \quad (4)$$

75 so an aligned neighborhood shields a site ($\lambda_i \rightarrow \lambda_0$) while a disordered neighborhood accelerates its
 76 degradation ($\lambda_i \rightarrow \lambda_0(1 + \beta)$); degradation is thus *autocatalytic in disorder*. Each diluted site is *repaired*
 77 ($g_0 \rightarrow 1$, θ_i reset toward 0) at rate R . The bare susceptibility k , drive G , geometric shielding ($1 - s u_s$),
 78 cooperativity h and gain β parameterize the flux. This state dependence couples the site process to the
 79 rotor field: the two cannot be treated separately, and detailed balance is broken, driving the joint system to
 80 a NESS.

81 The order parameter and the information variable are

$$82 \quad U_\chi = \left| N^{-1} \sum_i e^{i\theta_i} \right|, \quad I = 1 - U_\chi. \quad (5)$$

83 3. The order–entropy bridge

84 Fixing the mean resultant length and maximizing differential entropy over circular densities yields the
 85 von Mises law $vM(\kappa)$ [19, 20], with density

86

$$f(\theta) = \frac{e^{\kappa \cos \theta}}{2\pi I_0(\kappa)}, \quad A(\kappa) = \frac{I_1(\kappa)}{I_0(\kappa)}. \quad (6)$$

87 where I_0 and I_1 are modified Bessel functions of the first kind [21].

88 **Theorem 1 (Order–entropy bridge).** *For the von Mises family: (i) $U_\chi = A(\kappa)$ is a smooth strictly*
 89 *increasing bijection $[0, \infty) \rightarrow [0, 1)$; (ii) $S(\kappa) = \ln(2\pi I_0(\kappa)) - \kappa A(\kappa)$ satisfies $dS/d\kappa = -\kappa A'(\kappa) < 0$ for κ*
 90 *> 0 , hence $dS/dU_\chi = -\kappa$; (iii) S is a strictly decreasing invertible function of U_χ , with $S = \ln 2\pi$ at $U_\chi = 0$*
 91 *and $S \rightarrow -\infty$ as $U_\chi \rightarrow 1$; (iv) $H_\chi = S_{\max} - S = -\frac{1}{2} \ln(1 - U_\chi) + \frac{1}{2}(\ln \pi - 1) + o(1)$ as $U_\chi \rightarrow 1$.*

92 **Proof.** (i) The von Mises family is an exponential family in $\cos \theta$ with log-partition function $\ln(2\pi I_0(\kappa))$,
 93 so $A(\kappa) = d \ln I_0(\kappa)/d\kappa = E_\kappa[\cos \theta]$, and $A'(\kappa) = \text{Var}_\kappa(\cos \theta)$, which is strictly positive for every $\kappa \geq 0$
 94 because $\cos \theta$ is not almost surely constant under any von Mises law. The Bessel recurrences give the
 95 Riccati identity (see [21] for properties of the ratio A)

96

$$A'(\kappa) = 1 - \frac{A(\kappa)}{\kappa} - A(\kappa)^2, \quad (7)$$

97 and $0 < A(\kappa) < 1$ for $\kappa > 0$. With $A(0) = 0$ and the asymptotics $A(\kappa) = 1 - 1/(2\kappa) + O(\kappa^{-2})$ as $\kappa \rightarrow \infty$, A is a
 98 strictly increasing bijection $[0, \infty) \rightarrow [0, 1)$, which proves (i). (ii) Since $(d/d\kappa) \ln(2\pi I_0) = A$ and $(d/d\kappa)$
 99 $(\kappa A) = A + \kappa A'$, $dS/d\kappa = -\kappa A' < 0$; dividing by $dU_\chi/d\kappa = A'$ gives $dS/dU_\chi = -\kappa$. (iii) follows by the
 100 inverse-function theorem. (iv) $I_0(\kappa) \sim e^\kappa/\sqrt{2\pi\kappa}$ gives $S = \frac{1}{2} + \frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \kappa + o(1)$, and inverting $A(\kappa)$
 101 $\approx 1 - 1/(2\kappa)$ near $U_\chi \rightarrow 1$ ($\kappa \sim 1/[2(1 - U_\chi)]$) yields the stated tail. ■

102 The identity $dS/dU_\chi = -\kappa$ is exact: the marginal information gained per unit increase in directional order
 103 equals the concentration of the underlying law, so the last increments toward perfect order are
 104 logarithmically costly (Figure 1).

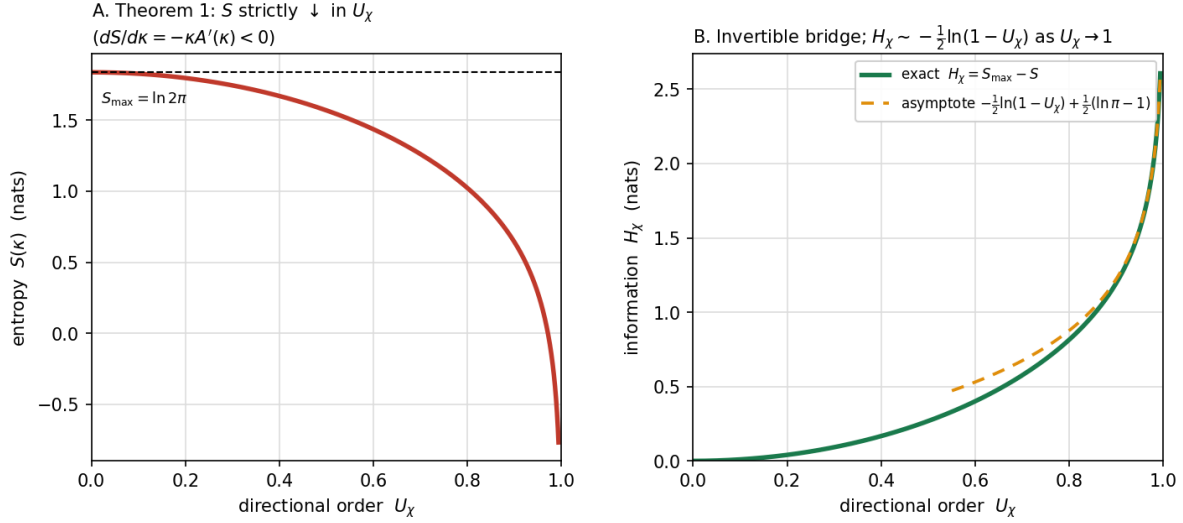


Figure 1. The order–entropy bridge (Theorem 1). (A) S is strictly decreasing in U_χ . (B) $H_\chi = S_{\max} - S$ is a strictly increasing invertible image of U_χ , with the derived logarithmic tail (dashed).

4. Mean-field reduction, feedback and bistability

4.1 The uncoupled baseline. Setting $\beta = 0$ removes the feedback: the integrity of each site is then an exact two-state Markov chain (intact $\rightleftharpoons$ diluted at rates λ_0, R), independent of the rotors. The intact fraction obeys $\dot{p} = R(1 - p) - \lambda_0 p$, so with $\tau = \lambda_0 t$ and $\rho = R/\lambda_0$,

$$\frac{dI}{d\tau} = (1 - I) - \rho I, \quad I^* = \frac{1}{1 + \rho}. \quad (8)$$

a globally stable fixed point approached at rate $(1 + \rho)$; the lattice $p(t)$ follows this master equation up to a discrete-time correction: the simulation updates all sites synchronously once per step, so its exact fixed point is $R/(\lambda_0 + R - \lambda_0 R)$ rather than $R/(\lambda_0 + R)$, a relative shift below $\lambda_0/(1 - \lambda_0) \approx 2\%$ that vanishes as $\lambda_0, R \rightarrow 0$ (it is checked to Monte Carlo accuracy in Appendix A) and U_χ is slaved to it (Figure 2). The linear baseline is recovered as the $\beta = 0$ limit.

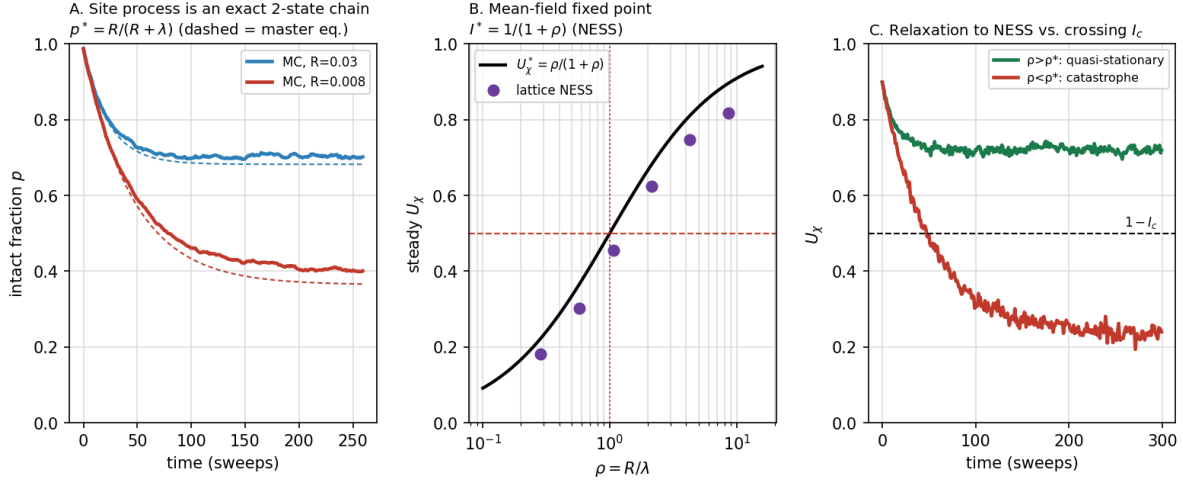


Figure 2. Uncoupled baseline ($\beta = 0$). (A) Lattice intact fraction follows the two-state master equation (dashed) up to the discrete-time correction of Section 4.1. (B) NESS order lies on $U_x^* = \rho/(1 + \rho)$. (C) Relaxation to the NESS for $\rho > \rho_c$ and crossing of I_c for $\rho < \rho_c$.

4.2 Nonlinear feedback and bistability. With order-dependent degradation, the annealed mean-field identification $\delta \doteq I$ closes the feedback loop and the information variable obeys the nonlinear ODE

$$\frac{dI}{d\tau} = \mu(I)(1 - I) - \rho I, \quad \mu(I) = 1 + \beta \frac{I^h}{K^h + I^h}. \quad (9)$$

Proposition 2 (Bistable NESS). *Steady states solve*

$$\rho = F(I) := \frac{\mu(I)(1 - I)}{I}. \quad (10)$$

For cooperative feedback ($h > 1$ and $\beta > \beta_c(h, K)$) F is non-monotonic, so there is an interval $\rho \in (\rho_-, \rho_+)$ on which the ODE has three fixed points: two stable — a maintained (low- I) branch and a collapsed (high- I) branch — separated by an unstable one. The endpoints $\rho_{\pm}$ are saddle-node bifurcations, and the system exhibits hysteresis: decreasing ρ from the maintained state, order persists until ρ_- , then collapses; increasing ρ from the collapsed state, order is not recovered until ρ_+ . For $(h, K, \beta) = (6, 0.5, 20)$ the bistable window is $\rho \in (3.9, 11.3)$.

Proof sketch. $F(I) = \mu(I)(1 - I)/I \rightarrow \infty$ as $I \rightarrow 0^+$, because $\mu(0) = 1$, and $F(I) \rightarrow 0$ as $I \rightarrow 1^-$. For $h > 1$ and large β the cooperative term contributes a rising factor that makes F non-monotonic, with a local minimum and a local maximum whose values are ρ_- (minimum) and ρ_+ (maximum). Between them three real roots exist; linearizing $G(I) = \mu(I)(1 - I) - \rho I$ gives stability $\text{sgn}(-G'(I^*))$, stable on the outer branches and unstable on the middle one. The saddle-node conditions $F'(I) = 0$ locate $\rho_{\pm}$. ■

Where the feedback is strong enough. Cooperativity is necessary. For $h \leq 1$, $F(I) = (1 - I)/I + \beta I^{h-1}(1 - I)/(K^h + I^h)$ is a sum of strictly decreasing functions, so the steady state is unique for every β . For $h > 1$ the threshold $\beta_c(h, K)$, the smallest gain at which F develops its pair of turning points, follows from the saddle-node conditions as in the companion order–repair analysis [22]. At $K = 0.5$ it is $\beta_c = 219, 50.7, 19.1, 7.73$ and 2.42 for $h = 1.2, 1.5, 2, 3$ and 6 , diverging as $h \rightarrow 1^+$. The parameters used throughout, $(h, K, \beta) = (6, 0.5, 20)$, lie at $\beta/\beta_c = 8.3$, where the window is $\rho \in (3.92, 11.30)$; it shrinks to a point as $\beta \rightarrow \beta_c$ and widens to $(4.57, 21.8)$ at $\beta = 40$ (`mf_threshold.py` in the study repository).

Figure 3 shows the S-shaped steady-state manifold, the bifurcation diagram with its hysteresis window, and two trajectories at a common ρ relaxing to distinct (maintained vs collapsed) NESS. Bistability is the qualitative signature of the feedback: it is absent in the $\beta = 0$ baseline and appears only because degradation is coupled to the order it destroys.

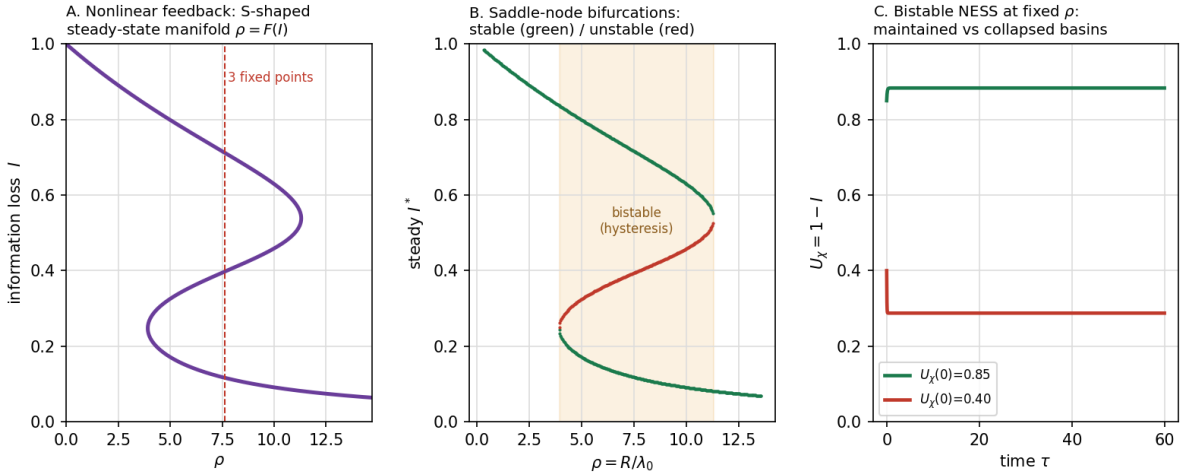


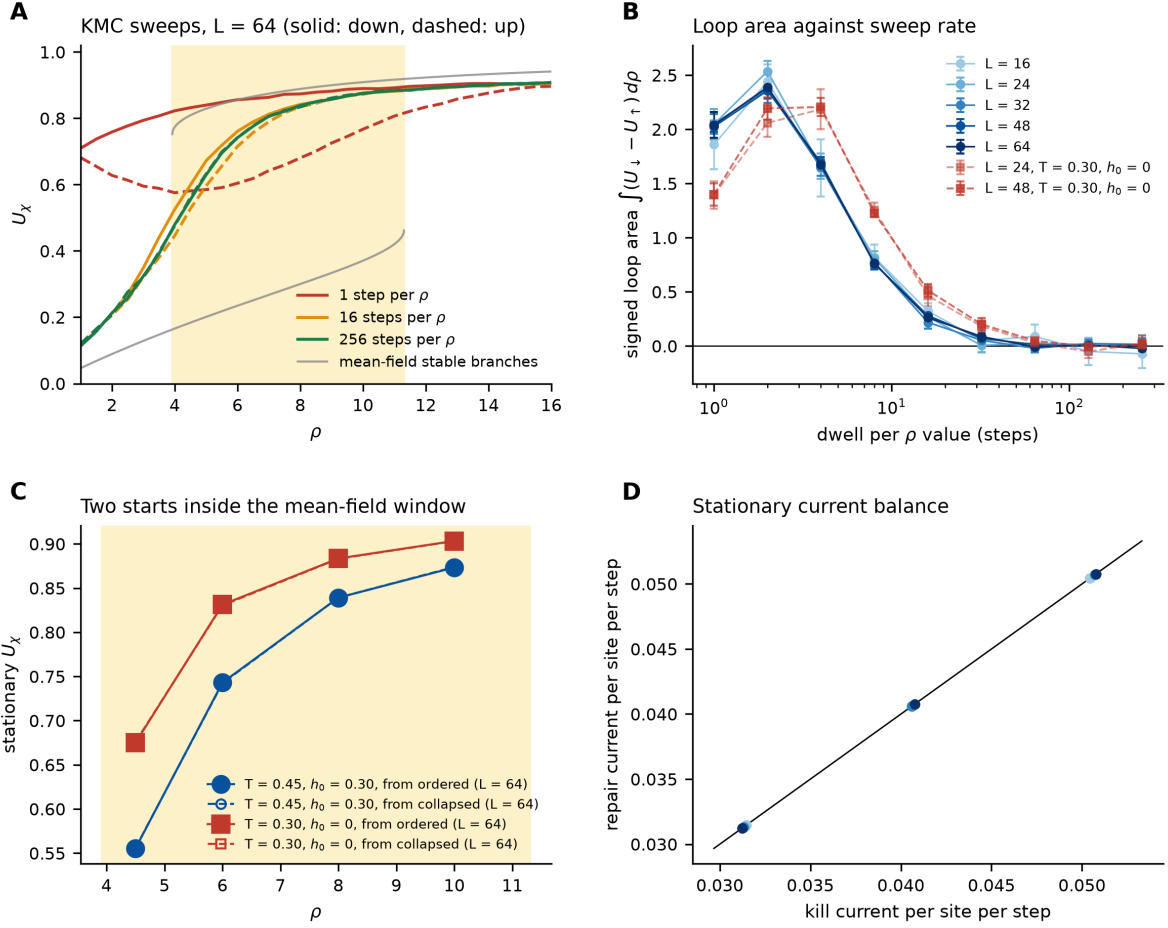
Figure 3. Nonlinear feedback $\rightarrow$ bistability (Proposition 2). (A) The steady-state manifold $\rho = F(I)$ is S-shaped, so a range of ρ admits three fixed points. (B) Bifurcation diagram: stable (green) and unstable (red) branches with two saddle-node points bounding a hysteresis window (shaded). (C) At fixed ρ inside the window, initial conditions flow to a maintained or a collapsed NESS.

4.3 Testing the bistability on the lattice. Proposition 2 is a statement about the reduced ODE, obtained under one approximation: the *annealed mean-field closure* $\delta \doteq I$, which replaces the local neighbourhood disorder δ_i of each site by the global average I . Whether its central prediction survives on the short-range lattice is a numerical question, and a hysteresis loop recorded at one sweep rate cannot settle it: a loop opens whenever the control parameter moves faster than the system relaxes, whether or not two stable states exist [23]. We therefore use kinetic Monte Carlo [24] to apply two tests that separate dynamic from static hysteresis, at the parameters of Proposition 2 ($T = 0.45, h_0 = 0.30$) and in a colder, field-free variant ($T = 0.30, h_0 = 0$) in which thermal fluctuations are weaker. The protocol, run table, engine verification and error analysis are given in Appendix A.

163 *Sweep-rate extrapolation.* The ratio ρ is stepped from 16 down to 1 and back on a 31-point grid with a
 164 dwell of $\tau = 1$ to 256 steps at each value, at $L = 16\text{--}64$ (8 independent runs per size and rate). The signed
 165 loop area $A = \int (U_{\chi\downarrow} - U_{\chi\uparrow}) d\rho$, which unlike the absolute area is unbiased under noise, peaks at 2.5 for
 166 dwells of two to four steps and falls steadily as the sweep slows, to 0.09 at $\tau = 32$ for $L = 64$ (Figure 4B).
 167 From $\tau = 64$ on it is consistent with zero: $A = -0.007$ (95% bootstrap interval -0.020 to 0.005) over the
 168 dwells $\tau = 64\text{--}256$ at $L = 64$, $|A| \leq 0.018$ at every size in both variants, and every interval contains zero. A
 169 bistable system keeps a finite loop as $\tau \rightarrow \infty$, bounded by its two saddle-nodes; for the mean-field
 170 reduction that quasi-static area is 4.30.

171 *Two starts at fixed ρ .* Inside the mean-field window ($\rho = 4.5, 6, 8$ and 10) each run starts either from the
 172 ordered state or from a collapsed one (95% of sites diluted, angles random) and is followed for 3000 steps
 173 ($L = 16\text{--}64$, 8 runs per start). The stationary order reached from the two starts agrees to within 0.004 in all
 174 40 combinations of ρ , L and variant, with the bootstrap interval of the difference containing zero in 35 of
 175 them (Figure 4C). The seed-averaged trajectories meet within 15–115 steps, and the meeting time does
 176 not grow with the number of sites from $N = 256$ to 4096 (Figure A1C), whereas escape from a metastable
 177 branch slows exponentially with N . The stationary distribution of U_χ is unimodal (bimodality coefficient $\leq$
 178 0.37, below the 5/9 of a uniform distribution [25]), and its integrated autocorrelation time is 6–11 steps,
 179 again independent of L (Figure A1B).

180 Both tests give the same answer. At these parameters the nearest-neighbour lattice has a single stationary
 181 state at each ρ , and the hysteresis it shows is dynamic: a lag behind a moving control parameter that
 182 vanishes in the quasi-static limit. The slow sweeps trace one continuous crossover (Figure 4A) across the
 183 range in which the reduction predicts two branches, and at $\rho = 6$, where the reduction has stable states at
 184 $U_\chi = 0.86$ and 0.23 , both starts settle at $U_\chi = 0.74$. Mean-field bistability does not survive on the lattice at
 185 these parameters, including the colder variant. The stationary currents nonetheless balance, $J_{\text{kill}} = J_{\text{rep}}$ to
 186 better than one part in 10^3 at every ρ and L (Figure 4D), so the lattice reaches a genuine NESS and the
 187 cycle current used in Section 6 is the measured one.



188

189 **Figure 4.** Kinetic Monte Carlo test of Proposition 2 on the nearest-neighbour lattice ($T = 0.45$, $h_0 = 0.30$ unless
190 stated). (A) Sweeps of ρ down (solid) and up (dashed) at $L = 64$, with 1, 16 and 256 steps per ρ value: the loop of
191 fast sweeps closes as the sweep slows; the mean-field window is shaded and its stable branches are grey. (B) Signed
192 loop area against dwell per ρ value for $L = 16$ – 64 and for the colder field-free variant (squares); mean $\pm$ 95%
193 interval over 8 runs. (C) Stationary order from an ordered and a collapsed start at fixed ρ inside the window ($L =$
194 64): the two coincide. (D) Stationary kill and repair currents per site and step at $\rho = 4.5, 6$ and 8 and $L = 16, 32, 64$;
195 the line is equality.

196 **4.4 Why the closure fails, and where it holds.** The reduction assumes that every intact site is degraded
197 at the rate $\mu(I)$ set by the global disorder. The lattice applies instead the average of μ over the local
198 disorders its intact sites actually see, $\bar{\mu} = \langle \mu(\delta_i) \rangle$, and the two differ in both directions (Figure 5A). Where
199 order is low the local disorder of intact sites stays far below the global one: at $\rho = 2$ the global I is 0.79
200 while intact sites see $\delta = 0.45$, so the lattice kills at $\bar{\mu} = 9.1$ against $\mu(I) = 19.8$. Part of this is a ceiling:
201 four rotors pointing in independent random directions give $\delta = 0.55$ on average, barely above $K = 0.5$, so
202 a nearest-neighbour sensor can never engage the switch fully. Where order is high the lattice kills faster
203 than the closure assumes: at $\rho = 6$, $\bar{\mu} = 2.47$ against $\mu(I) = 1.37$, and above even $\mu(\delta) = 1.02$, the rate at the
204 mean local disorder. That last inequality is the signature of averaging a steep, convex switch over a broad

205 distribution of δ_i : the few sites in disordered pockets are killed at nearly the full rate even in an ordered
206 lattice. Together the two effects flatten the effective feedback, and a flattened μ cannot bend F into the S-
207 shape that Proposition 2 requires. Both shrink as the neighbourhood that a site senses grows, since the
208 average over n rotors narrows the distribution of δ_i and raises its ceiling (to 0.82 for $n = 24$ and 0.90 for n
209 $= 80$).

210 A control experiment tests this directly. Keeping the rotor dynamics, the repair process and every
211 parameter fixed, we let the kill rate respond to the disorder of a larger block of n surrounding sites ($n = 8$,
212 $24, 80$) or to the global disorder $1 - U_\chi$, which imposes the mean-field closure on the lattice itself ($L = 48$,
213 4 runs per case; Figure 5B–D). With $n = 4$ the slow-sweep loop vanishes (area 0.01 at $\tau = 256$) and with n
214 $= 8$ it nearly does (0.03); the two starts agree at every ρ . With $n = 24$ a small loop remains (0.21, still
215 shrinking between $\tau = 64$ and 256) while the starts still agree at the values of ρ tested. With $n = 80$ the
216 loop has area 1.00 and at $\rho = 6$ the ordered start holds $U_\chi = 0.85$ while the collapsed start stays at 0.39
217 throughout the run. With global sensing the two starts are separated at $\rho = 4.5$ (0.82 against 0.28) and 6
218 (0.85 against 0.37), and the slow sweep collapses at $\rho \approx 3.7$, close to the mean-field saddle-node $\rho_- =$
219 3.92 , but recovers already at $\rho \approx 7.5$, below $\rho_+ = 11.30$: at finite N fluctuations let the collapsed state
220 escape before its own saddle-node, so the loop (area 1.93) is about half the quasi-static mean-field value
221 4.30. In both bistable cases the two branches persisted for the whole 2000-step runs, some 200
222 autocorrelation times; at a single size we do not establish how these lifetimes grow with N .

223 The bistability of Proposition 2 is therefore not an artefact of the reduction: it is realized on the lattice
224 when degradation responds to order averaged over a large enough region, and it is the mean-field limit of
225 that case. It is destroyed when degradation responds only to the nearest neighbours; at the parameters
226 studied the crossover lies between $n = 8$ and $n = 80$ sensed sites. The physical content of Proposition 2 is
227 accordingly conditional on the range over which the degrading flux senses disorder, and that range, not
228 only the gain β and the cooperativity h , is a control parameter of the maintained state.

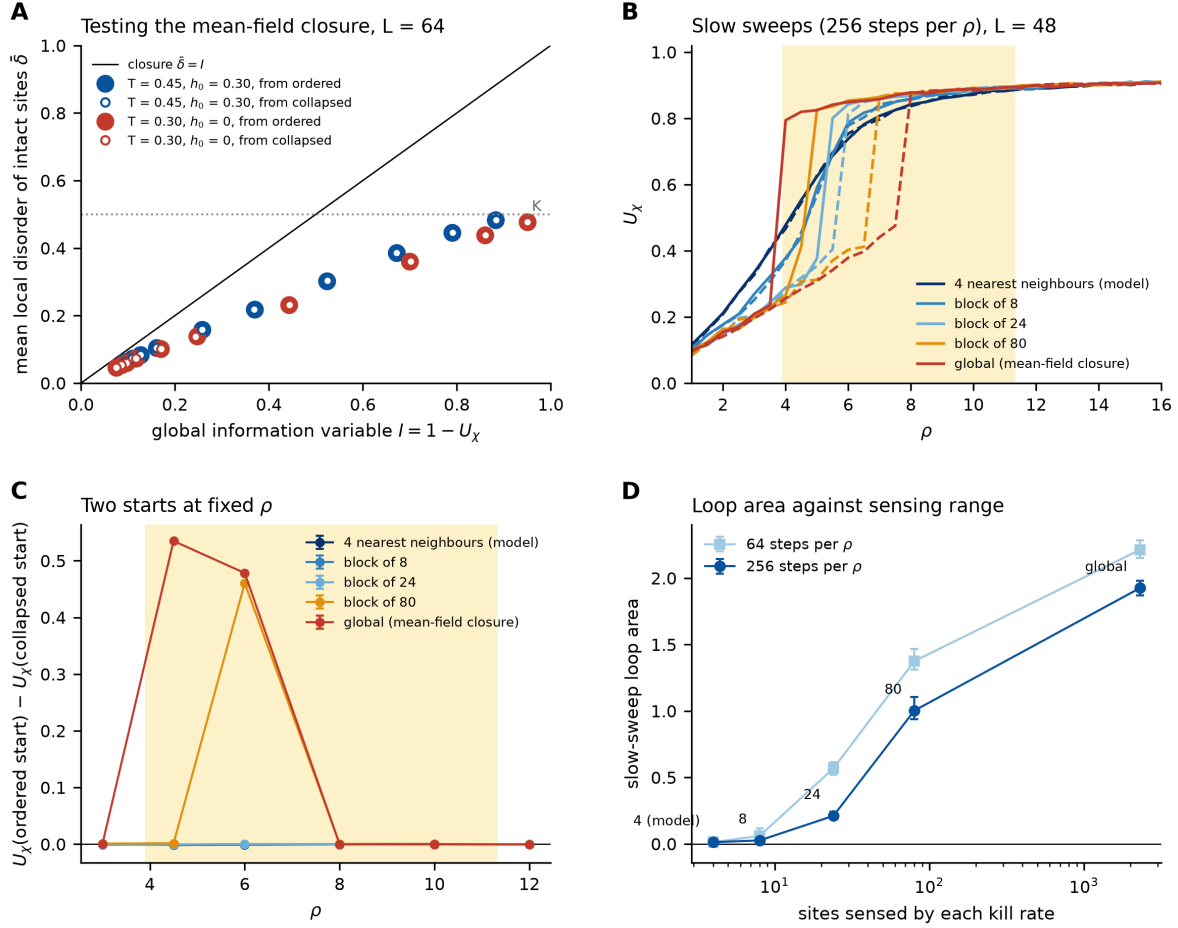


Figure 5. Why nearest-neighbour feedback is not bistable, and the sensing range that restores bistability. (A) Mean local disorder of intact sites $\bar{\delta}$ against the global $I = 1 - U_\chi$ in the stationary lattice ($L = 64$; filled: ordered start, open: collapsed start; both parameter sets); the line is the closure $\bar{\delta} = I$ and the dotted line marks K . (B) Slow sweeps (256 steps per ρ value) at $L = 48$ with the kill rate sensing the 4 nearest neighbours (the model), blocks of 8, 24 and 80 sites, or the global disorder; solid down, dashed up; the mean-field window is shaded. (C) Difference in stationary order between the ordered and the collapsed start at fixed ρ , with 95% intervals. (D) Slow-sweep loop area against the number of sites sensed, at 64 and 256 steps per ρ value.

5. The field-free transition is BKT-driven

At $h_0 = 0$ the maintained phase requires the intact sub-lattice to sustain the quasi-long-range order of the two-dimensional XY model [12]. We locate the field-free transition of the quenched problem, in which a fixed fraction f of sites is diluted to $g = 0$, by finite-size scaling at $T = 0.45$ with $L = 16, 24, 32, 48, 64$ and 96 and eight independent disorder realizations per size and dilution (Appendix A), measuring the Binder cumulant U_4 and the helicity modulus (spin stiffness)

$$Y = \frac{1}{L^2} \left\langle \sum_b J \cos \Delta \theta \right\rangle - \frac{1}{L^2 T} \left\langle \left(\sum_b J \sin \Delta \theta \right)^2 \right\rangle, \quad (11)$$

averaged over the two lattice directions. The Nelson–Kosterlitz criterion [26] places the BKT transition where Y crosses the universal line $2T/\pi$. Dilution lowers the effective coupling and with it the effective transition temperature, and Y crosses $2T/\pi$ at $f_{KT}(L) = 0.220$ for $L = 16$, falling slowly to 0.210 at $L = 96$, with bootstrap intervals over realizations within ± 0.005 for $L \geq 48$ (Figure 6A, B). The drift is small and of the logarithmic kind expected at a BKT point, and two extrapolations agree: a linear fit in $1/\ln^2 L$ gives $f_{KT}(\infty) = 0.205$ (95% interval 0.193 – 0.216), and a Weber–Minnhagen fit [27] of $Y_L = (2T/\pi)[1 + 1/(2 \ln L + C)]$ to the sizes $L \geq 24$ gives $f_{KT} = 0.205$ (0.198 – 0.213) with $\chi^2 = 0.75$ for 4 degrees of freedom. Both lie far below the square-lattice site-percolation value $1 - p_c \approx 0.407$ [28]: the intact sub-lattice still percolates geometrically at f_{KT} , so order is lost to thermal (BKT) fluctuations rather than to geometric disconnection. The Binder cumulant (Figure 6C) stays at its ordered value up to $f \approx 0.23$ and falls between $f = 0.26$ and 0.35 , again well before percolation. The magnitude is plausible: the clean model has $T_{BKT} \approx 0.893$ [29], twice the temperature used here, and a linear interpolation of $T_{BKT}(f)$ between that value and zero at percolation, a heuristic only, would bring the transition down to $T = 0.45$ at $f \approx 0.20$.

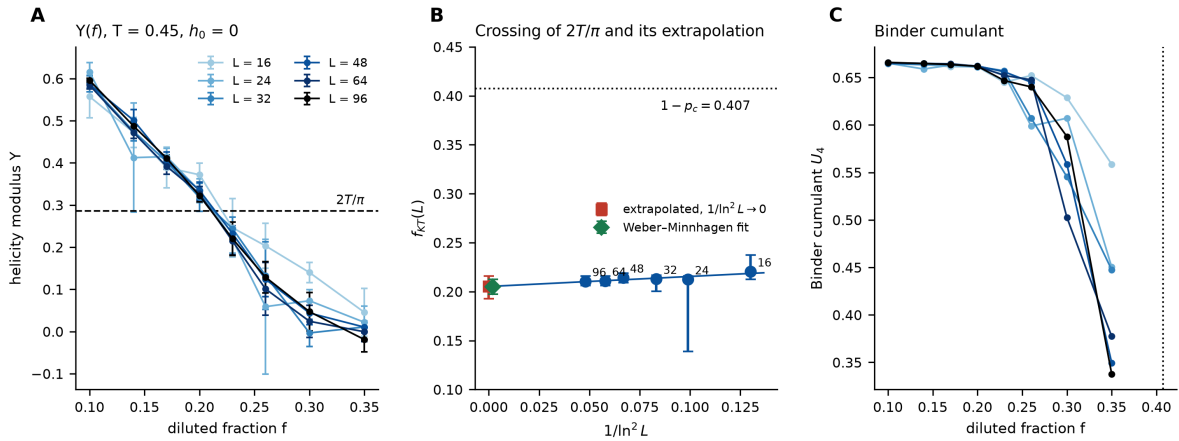


Figure 6. Finite-size scaling of the field-free ($h_0 = 0$) transition under quenched dilution ($T = 0.45$; $L = 16$ – 96 ; 8 disorder realizations per point; error bars are 95% intervals). (A) Helicity modulus $Y(f)$ and the Nelson–Kosterlitz line $2T/\pi$ (dashed). (B) Crossing points $f_{KT}(L)$ with bootstrap intervals against $1/\ln^2 L$, the linear extrapolation (square) and the Weber–Minnhagen estimate (diamond); the dotted line is the site-percolation value $1 - p_c \approx 0.407$. (C) Binder cumulant $U_4(f)$; the dotted line marks $1 - p_c$.

6. Entropy production and the cost of maintenance

The maintenance loop is a driven cycle in the joint (integrity, orientation) space: a site is killed (order $\rightarrow$ disorder) through the flux channel and repaired (disorder $\rightarrow$ order) through the maintenance channel. Representing the two mesostates O (ordered/intact) and D (disordered/diluted) with two *parallel driven edges* — a kill edge (O $\rightarrow$ D rate λ , reverse ε) and a repair edge (D $\rightarrow$ O rate R , reverse ε) — gives a Schnakenberg network [30] with cycle affinity

$$A = \ln\left(\frac{\lambda R}{\varepsilon^2}\right), \quad (12)$$

and NESS current J . The entropy production rate is

$$\sigma = J \cdot A \geq 0, \quad J = \lambda U_\chi^* \quad (\text{maintained regime}). \quad (13)$$

where $\varepsilon > 0$ is a small reverse rate rendering the cycle affinity finite (a modeling choice; the strictly irreversible limit $\varepsilon \rightarrow 0$ gives $\sigma \rightarrow \infty$), and the stationary current J is the one measured in the stationary lattice (Section 4.3, Figure 4D). Equality $\sigma = 0$ holds only when the drive vanishes. Figure 7 shows $\sigma(\rho) > 0$ wherever the cycle is driven forward, $\lambda R > \varepsilon^2$, i.e. $\rho > \varepsilon^2/\lambda^2$ (the whole plotted range, since ε is small), confirming that the stationary state is a genuine NESS held away from thermal equilibrium, and $\sigma \rightarrow 0$ as ρ falls to ε^2/λ^2 , which tends to $\rho \rightarrow 0$ as $\varepsilon \rightarrow 0$, where maintenance ceases and the system relaxes by passive decay. Because $J = \lambda U_\chi^*$, the maintenance throughput — and hence the modeled entropy production — increases with the maintained directional order; because directional order is monotonically related to configurational information (Theorem 1), this establishes a thermodynamic cost associated with maintaining information-bearing order, though the two are not generally proportional ($dS/dU_\chi = -\kappa$ is not constant). This makes explicit the thermodynamic price of information maintenance and situates the model squarely within stochastic thermodynamics [16].

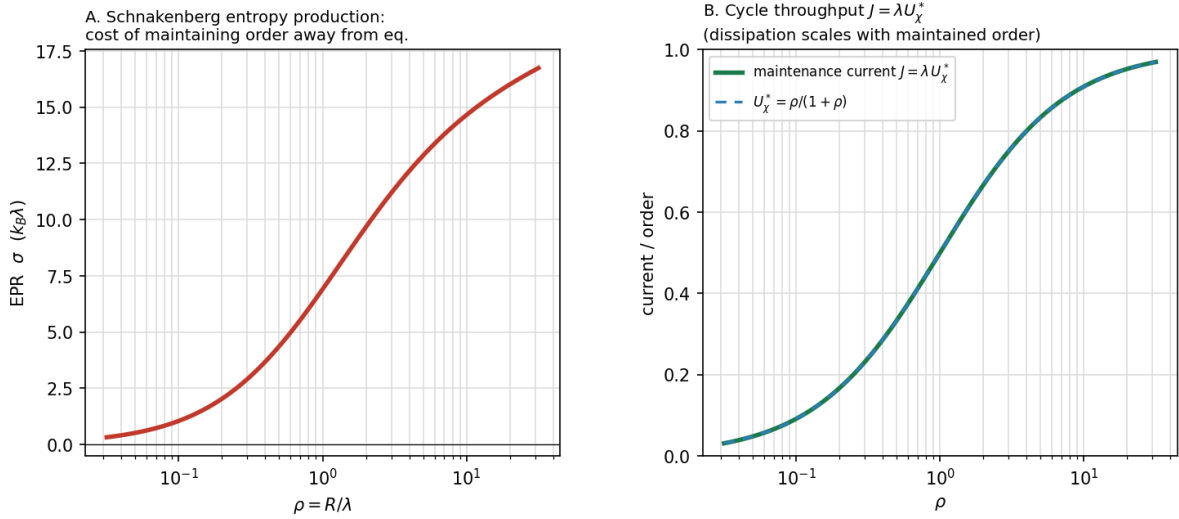


Figure 7. Schnakenberg entropy production. (A) $\sigma(\rho) > 0$ for $\rho > \varepsilon^2/\lambda^2$ (genuine NESS) and $\sigma \rightarrow 0$ as $\rho \rightarrow \varepsilon^2/\lambda^2$ (passive decay). (B) The cycle throughput $J = \lambda U_\chi^*$ tracks the maintained order, so dissipation grows with the order held.

288 7. Generality and instances

289 The construction assumes only three ingredients: a directional order parameter admitting a maximum-
290 entropy law (Theorem 1), an order-dependent flux that removes ordering couplings (the feedback of
291 Section 4), and a repair process that restores them. Systems satisfying these ingredients are expected to
292 exhibit analogous feedback-driven behavior, although the quantitative thresholds and bifurcation structure
293 will depend on their microscopic dynamics — real systems may carry additional degrees of freedom
294 (spatial correlations, memory, non-Markovian repair) that this minimal model does not capture. Instances
295 include the orientational order of a polymer film degraded by ultraviolet chain scission and restored by
296 annealing or self-healing chemistry; magnetic storage domains under thermally activated flips with active
297 error-correcting rewrites (repair); liquid-crystal directors under photochemical damage; and the chiral
298 order of a self-assembled molecular monolayer under non-enzymatic attack with turnover, the applied
299 problem from which this model was abstracted [9]. In each, u_s encodes the geometry throttling the flux
300 and ρ measures repair relative to shielded, order-dependent degradation. We emphasize that these are
301 candidate realizations of the same qualitative mechanism, not demonstrated physical instances: the
302 present study is analytic and computational, with no experimental data used (Data and code availability),
303 and the framework should be read as a minimal model that may be applicable to systems with order-
304 dependent degradation and active repair rather than as a validated description of any one of them.

305 7.1 A falsifiable experimental signature

306 The mechanism makes a concrete prediction that does not require the rotor language of the model, and
307 Sections 4.3 and 4.4 sharpen it. In any driven, actively maintained, orientationally ordered medium (a
308 magnetic-domain array, a self-healing polymer film, an aligned colloidal or active-particle ensemble),
309 sweep the ratio of repair to degradation down from a well-ordered state and back up at several sweep
310 rates, holding all other conditions fixed. Every such medium shows a loop when the sweep is fast, so the
311 discriminating observable is the loop area extrapolated to zero sweep rate, supported by the dependence
312 of the final state on the initial one at fixed ρ . A finite static loop, with order lost below a threshold ρ_- and
313 not regained until $\rho_+ > \rho_-$ however slowly the sweep is made, indicates feedback acting over an extended
314 range, as in Proposition 2. A loop that closes as the sweep slows, as on the nearest-neighbour lattice,
315 indicates feedback that is local, too weak ($\beta < \beta_c$) or not cooperative ($h \leq 1$); uncoupled two-state
316 degradation shows only the trivial lag of Section 4.1. Where the spatial range over which degradation
317 responds to order can be varied, for example through the diffusion length of a degrading species, the
318 crossover of Section 4.4 is itself testable. This gives Sections 3–6 a shared empirical target across the
319 candidate systems of this section.

8. Status of the claims and approximations

For transparency at the level expected in theoretical statistical physics, we state each principal claim, its logical status, and its support.

Table 1. Status of the principal claims, the logical status of each and its support.

Claim	Status	Support
Order–entropy bijection, $dS/dU_\chi = -\kappa$, tail law (Thm 1)	Proved	Complete analytic proof; Fig. 1
Linear reduction $\Gamma^* = 1/(1 + \rho)$ at $\beta = 0$	Exact (continuous time)	Two-state master equation; lattice agrees up to the discrete-time correction, Fig. 2, App. A
Bistability and saddle-nodes of the reduction (Prop. 2); threshold $\beta_c(h)$, none for $h \leq 1$	Exact for the mean-field ODE	Analytic; Fig. 3
Bistability on the nearest-neighbour lattice	Not realized at the parameters studied	Zero quasi-static loop area, converging starts, N-independent meeting times; Fig. 4, Fig. A1
Bistability with extended sensing (80 sites, global)	Observed numerically, one size	Finite slow-sweep loop, persistent split between starts; Fig. 5
Non-equilibrium steady state ($J_{\text{kill}} = J_{\text{rep}}$)	Validated numerically	Relative current imbalance below 10^{-3} ; Fig. 4D
BKT-driven field-free transition, $f_{\text{KT}} \approx 0.205 < 1 - p_c$	Validated (FSS, $L = 16-96$)	Helicity modulus vs $2T/\pi$, two extrapolations, Binder cumulant; Fig. 6
Schnakenberg EPR $\sigma(\rho) > 0$ for $\rho > \varepsilon^2/\lambda^2$	Analytic on a modeled cycle	$\sigma(\rho)$, Fig. 7; current measured, Fig. 4D

The framework rests on three explicitly identified approximations. (1) The *annealed mean-field closure* $\delta \doteq I$ underlies Proposition 2. It is now tested directly (Section 4.4): it fails for nearest-neighbour sensing, where it misstates the effective feedback in both directions, and becomes accurate as the sensing range grows; whether it becomes exact in a limit in which the range grows with N is open problem (a). (2) The *slaving* $U_\chi \doteq p$ in the $\beta = 0$ baseline holds up to a boundary-layer correction controlled by J_0 and h_0 . (3) The *finite reverse rate* ε in the Schnakenberg cycle regularizes the affinity; the strictly irreversible limit is singular. No other claim relies on an unstated approximation.

9. Open problems

(a) *Range of the feedback and the mean-field limit*: establish rigorously the scaling limit of the coupled rotor/integrity process and the conditions under which the nonlinear mean-field ODE, including its bistability, is exact, for instance in a Kac-type limit in which the sensing range grows with N [18]; locate the critical sensing range as a function of (β, h, K, T) ; and determine whether any region of nearest-neighbour parameters, such as much larger β or smaller K , is bistable.

(b) *First passage and large deviations*: in the bistable regime of extended sensing, characterize the distribution of the collapse time and the large-deviation rate for noise-induced transitions between the coexisting NESS, including how the lifetimes scale with N and with the sensing range.

(c) *Precise BKT determination*: the Nelson–Kosterlitz analysis here ($L \leq 96$) gives $f_{KT} \approx 0.205$; larger sizes, temperature scans along the dilution line and a test of the integral and detailed fluctuation theorems for the entropy production of Section 6 across the bifurcation remain.

10. Conclusion

Making degradation depend on the order it destroys turns a passive decay problem into a genuine non-equilibrium one. In mean field the maintained lattice is governed by a single ratio ρ through a nonlinear law that is bistable above a computed feedback threshold. On the lattice the same feedback is bistable only if degradation senses disorder over an extended region: with nearest-neighbour sensing the broad distribution of local disorder averages the switch away, the steady state is unique, and the hysteresis seen at finite sweep rates is dynamic and vanishes in the quasi-static limit. The range of the feedback thus joins its gain and cooperativity as a control parameter, and the loop area extrapolated to zero sweep rate is the observable that distinguishes the two regimes. The field-free loss of order is BKT-driven, at $f_{KT} \approx 0.205$, far below percolation, and the stationary state carries a positive entropy production that quantifies the cost of holding directional order, equivalently configurational information, away from equilibrium; the order–entropy bridge ties that information to the concentration of the maximum-entropy law through the exact relation $dS/dU_\chi = -\kappa$. The framework is substrate-free, packaging a family of maintenance-versus-decay problems into one reduced, dissipative law together with a lattice criterion for when that law holds.

Author contributions (CRediT)

Leon Sandler: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this revision the author used Claude (Anthropic) to assist with typesetting the equations in the journal's format, writing and testing the Monte Carlo and analysis code of Sections 4.3–5

and Appendix A to the author's specification, checking the reference list against Crossref records and language editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Data and code availability

All Monte Carlo results of this version, the engine, its verification suite, the analysis scripts that turn raw runs into every number and figure of Sections 4.3, 4.4 and 5 and of Appendix A, and the raw run outputs are openly available at <https://github.com/sandlerleon/rotational-lattice-maintenance>, archived at <https://doi.org/10.5281/zenodo.23130923>. Each run is reproducible on its own from its seed code (Appendix A). The figures of Sections 3, 4.1, 4.2 and 6 were produced with the open-source MyUncle framework (<https://github.com/sandlerleon/MyUncle>, <https://doi.org/10.5281/zenodo.21223569>), whose update rules the new engine reproduces exactly. This manuscript is archived as a preprint at <https://doi.org/10.5281/zenodo.21210708>. No experimental data were used.

Appendix A. Monte Carlo protocol and error analysis

A.1 Engine and verification. One step consists of two checkerboard Metropolis sweeps of the rotors at fixed integrity, with proposals uniform on $(-\pi, \pi]$, followed by one synchronous site update: each intact site is diluted with probability λ_i , then each diluted site is repaired with probability R and its rotor reset to $\theta = 0$. One step is the unit of time throughout. The engine restates the update rules of the MyUncle framework used for Figures 2, 3 and 7 and, before any result was computed, passed the checks of its test suite: it reproduces MyUncle trajectories bit for bit for the same seed at two parameter sets; at $\beta = 0$ the intact fraction relaxes to the discrete two-state fixed point $R/(\lambda_0 + R - \lambda_0 R)$ (0.7606 measured against 0.7614 exact, and 0.7500 for the continuous-time value); $R = 0$ never repairs and $\lambda_0 = 0$ never kills; the helicity modulus of the ordered undiluted lattice equals 1; and the block and global sensing options of Section 4.4 average exactly the intended sites.

A.2 Seeding and error estimates. Every run is seeded from a NumPy SeedSequence of a master seed (20261003) and a code naming its study, parameter set, size, control value, replicate and initial state, so runs are statistically independent and each can be repeated alone. Interval estimates are 95% percentile intervals from 2000 bootstrap resamples over independent runs or disorder realizations [31], so that variation between runs, not correlation within a run, sets the error.

A.3 Equilibration and autocorrelation. Integrated autocorrelation times are estimated with the automatic window $W \geq 6\tau_{\text{int}}$ [32]. In the stationary coupled lattice $\tau_{\text{int}}(U_\chi) = 6\text{--}11$ steps and τ_{int} of the intact fraction 9–16 steps for $L = 16\text{--}64$ and $\rho = 4.5\text{--}8$ (Figure A1A, B), with no growth with L . The 2000-step

discard of the stationary runs is therefore more than $150\tau_{\text{int}}$, and the second-half averages of the 3000-step two-start runs begin after 1500 steps, more than $100\tau_{\text{int}}$. In the quenched runs of Section 5, 2000 equilibration sweeps precede 4000 measurement sweeps; the longest τ_{int} of Y at any size and dilution is 69 sweeps, so each realization contributes at least 29 effectively independent samples, and the error of each point is in any case taken from the spread over the eight disorder realizations.

A.4 Run table. Table A1 lists every study.

Table A1. Monte Carlo runs. Times are in steps (coupled lattice) or sweeps (quenched lattice); “runs” are independent replicates per parameter combination.

Study (section)	Sizes L	Control values	Runs	Length; discard; sampling
Sweep rate (4.3)	16, 24, 32, 48, 64; colder variant 24, 48	$\rho = 16 \rightarrow 1 \rightarrow 16$, 31 values; dwell $\tau = 1\text{--}256$ (9 values)	8	200-step settle at $\rho = 16$, then 62τ ; U_χ at the end of each dwell
Two starts (4.3)	16, 24, 32, 48, 64	$\rho = 4.5, 6, 8, 10$; both parameter sets	8 per start	3000; first half; every 5
Stationary (4.3, A.3)	16, 32, 64	$\rho = 4.5, 6, 8$	4	12 000; 2000; every step
Closure (4.4)	64	$\rho = 1\text{--}16$ (10 values); both parameter sets and starts	4	800 + 1200; 800; every 4
Sensing range (4.4)	48	block of 8, 24, 80 sites, global; sweeps $\tau = 64, 256$; two starts at ρ $= 3\text{--}12$	4	sweeps as above; two starts 2000, first half, every 5
Quenched dilution (5)	16, 24, 32, 48, 64, 96	$f = 0.10\text{--}0.35$ (8 values), $T = 0.45$, $h_0 = 0$	8 realizations	2000 + 4000 sweeps; 2000; every 4

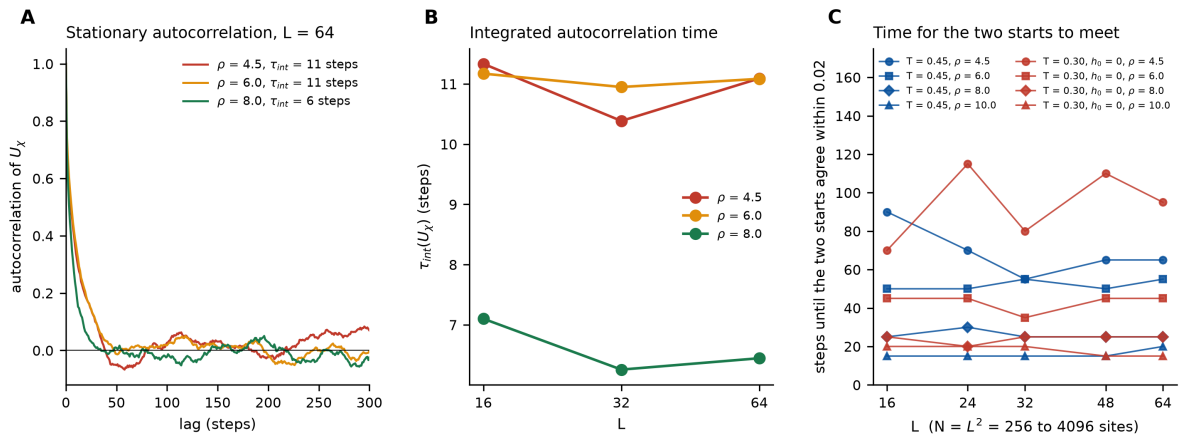


Figure A1. Monte Carlo diagnostics. (A) Stationary autocorrelation of U_χ at $L = 64$ for three values of ρ , with the integrated autocorrelation time. (B) $\tau_{\text{int}}(U_\chi)$ against L : no critical slowing down. (C) Time for the seed-averaged trajectories from the ordered and the collapsed start to agree within 0.02 (20-step moving average), against the number of sites, for both parameter sets and four values of ρ inside the mean-field window: it does not grow with N .

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